

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.6

13C NMR SPECTROSCOPY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.6: 13C NMR Spectroscopy
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.6

Q1. 1. (a) Predict the number of peaks in the ^{13}C NMR spectrum of (i) propan-1-ol and (ii) propan-2-ol.

[2 Marks]

WCH14/01/Jun22/Q9(a)

Q2. (b) A compound with molecular formula $\text{C}_4\text{H}_8\text{O}_2$ has 3 peaks in its ^{13}C NMR spectrum at $\delta = 20$ ppm, 60 ppm, and 170 ppm. Deduce its structure.

[3 Marks]

WCH14/01/Jun22/Q9(b)

SECTION B: WORKED MARK SCHEME — 15E.6

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Predict the number of peaks in the ^{13}C NMR spectrum of (i) propan-1-ol and (ii) propan-2-ol. 1. (i) Propan-1-ol: 3 peaks (3 non-equivalent C environments). 2. (ii) Propan-2-ol: 2 peaks (symmetrical, two identical CH_3 carbons).	[2]
Q2	Q2: (b) A compound with molecular formula $\text{C}_4\text{H}_8\text{O}_2$ has 3 peaks in its ^{13}C NMR spectrum at $\delta = 20$ ppm, 60 ppm, and 170 ppm. Deduce its structure. 1. Peak at 170 ppm = carbonyl carbon of ester/carboxylic acid ($\text{C}=\text{O}$). 2. Peak at 60 ppm = carbon attached to oxygen ($-\text{O}-\text{CH}_2-$). 3. Structure = ethyl ethanoate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$.	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15E.6**FAQ #1: How do I determine the number of ^{13}C NMR peaks?****Category: Symmetry & Environments****COMMON EXAMINER TRAP / MISTAKE:**

Counting total carbon atoms instead of carbon environments.

MODEL ANSWER / EXAMINER EXPECTATION:

Look for molecular symmetry. Carbon atoms in identical chemical environments give a single peak. Asymmetric molecules give one peak per carbon atom.